인공신경망 포텐셜: 실전 이론과 응용

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참고자료: Thesis by 유동선 박사 (SDC)

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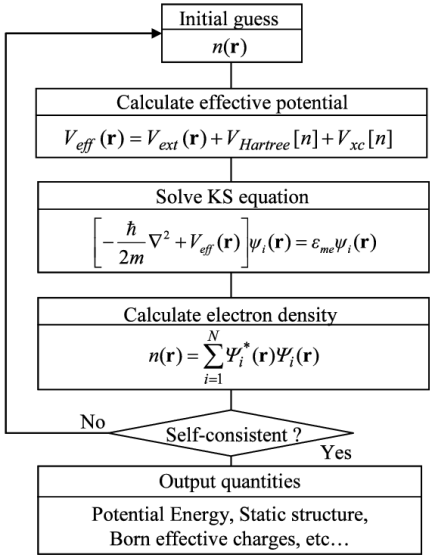
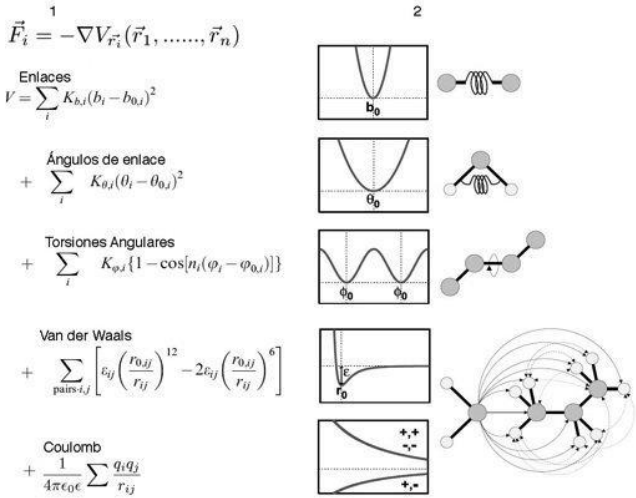
Methods for atomistic simulations

First-principles calculation



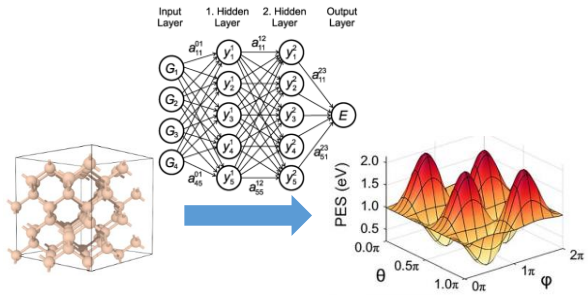
- Gives accurate and reliable results
- High computational cost

Classical force-field



- Fast but accuracy is limited
- Developed only for a few materials
- Difficult to develop models for new materials and complicated bonding natures

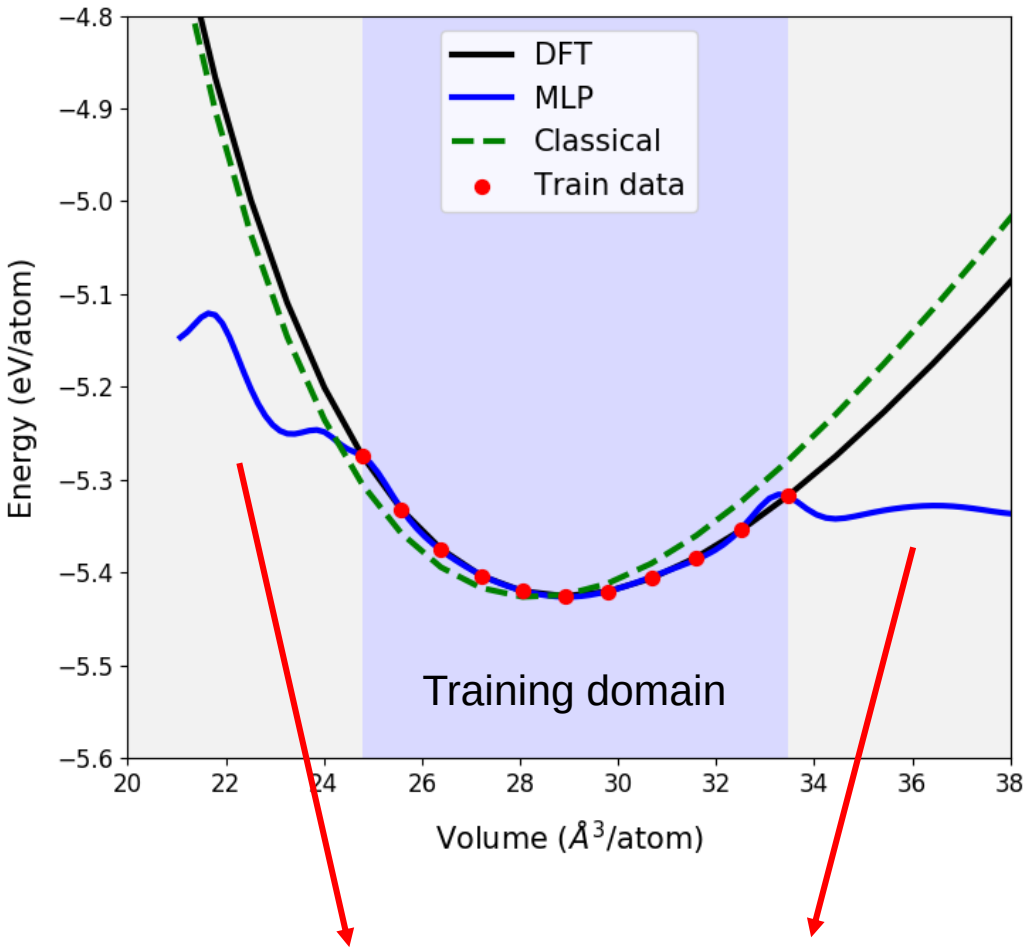
Machine learning potential (MLP)



- Learn structure-energy relation with machine learning models (flexible functional form)
- Gives accurate results and fast

DFT vs. classical potential vs. MLP

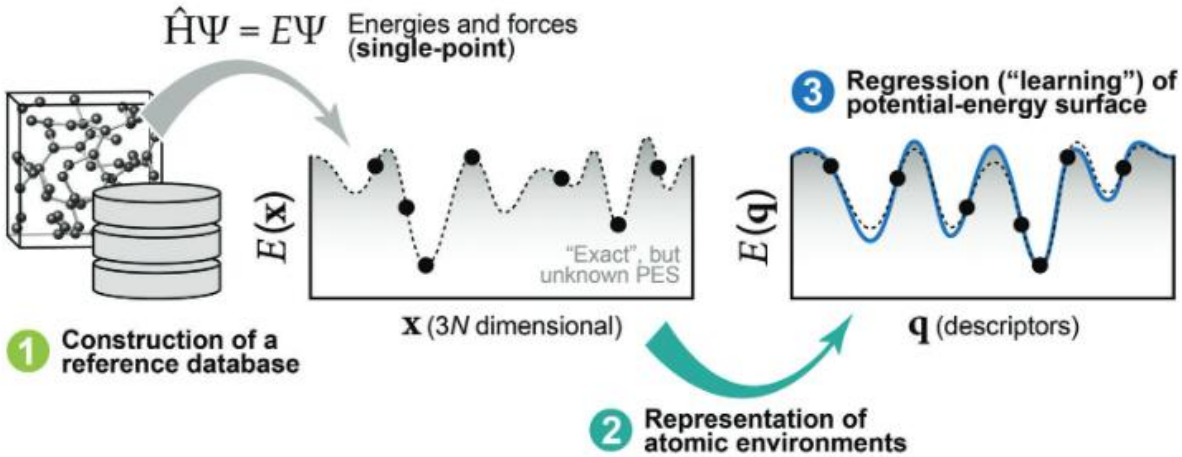
E-V curve for Si



Transferability is poor in extrapolation

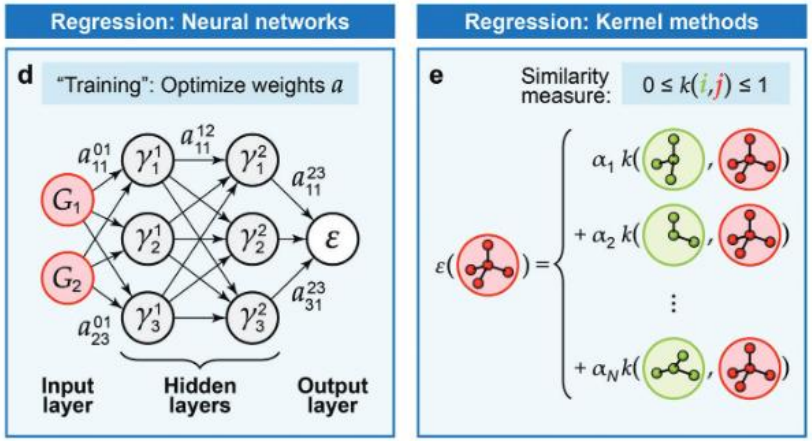
Machine-learning potential

V. L. Deringer, M. A. Caro and G. Csányi, *Adv. Mater.*, 1902765 (2019).



Three elements of MLP

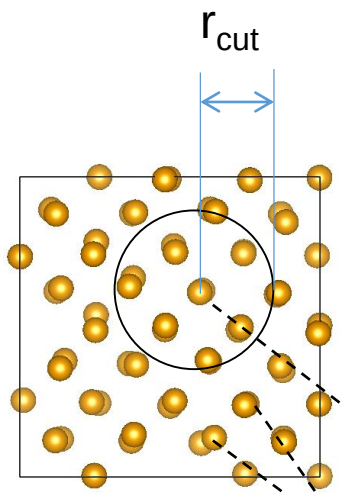
1. Training set
2. Descriptor
3. Model



- Neural network potential (NNP) : PRL **98**, 146401 (2007).
- Gaussian approximation potential (GAP) : PRL **104**, 136403 (2010)

High-dimensional neural network potential

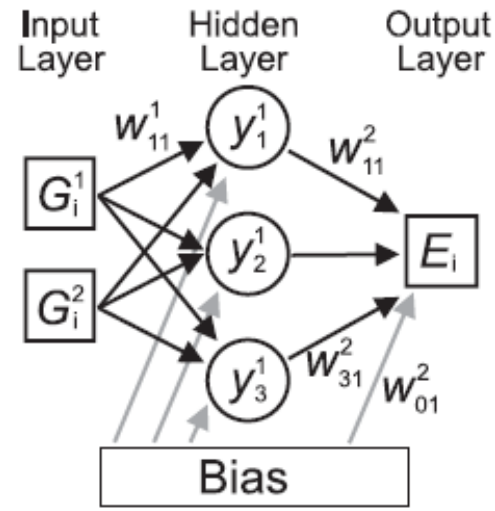
Behler and Parrinello, PRL (2007)

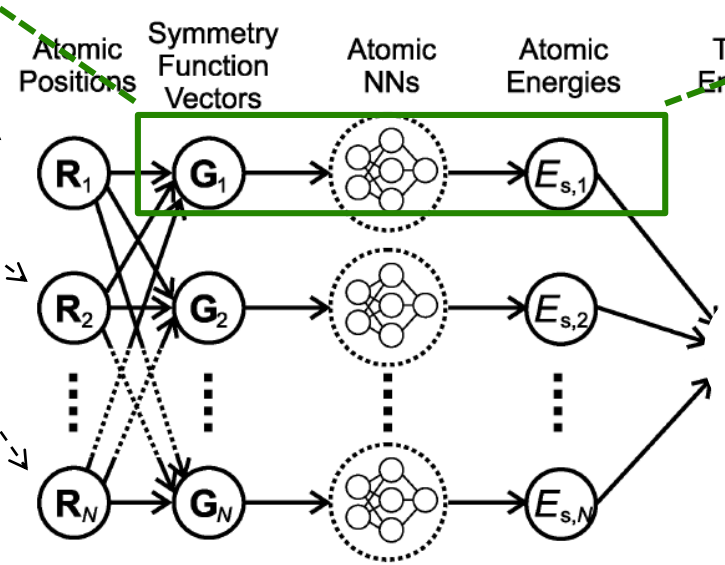


Atomic NN

- Feedforward NN

Local environment encoded by symmetry function vectors {**G**}

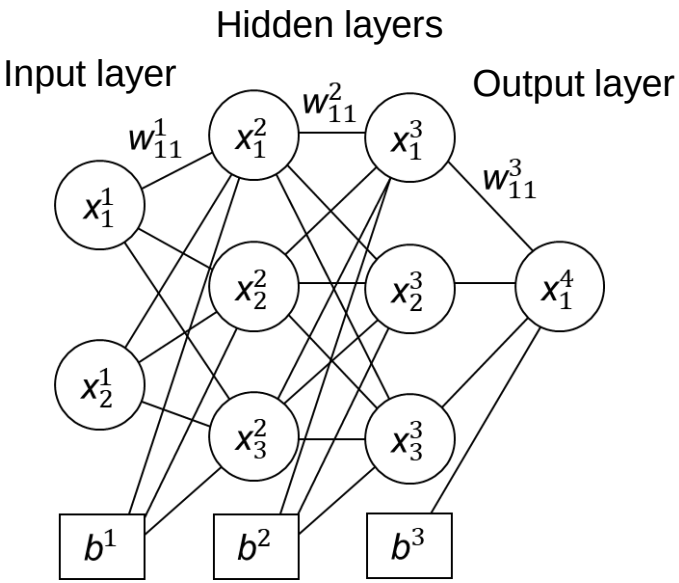


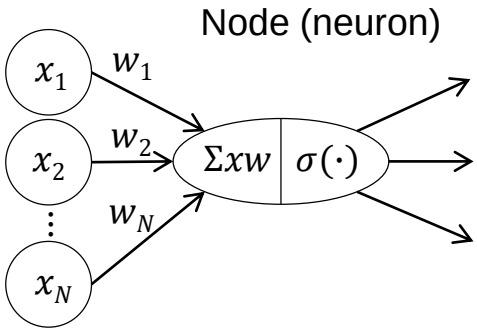


$$E_{\text{tot}} \approx \sum_i^{\text{atom}} E_{\text{at}}(\mathbf{R}_i; r_{\text{cut}})$$

$$= \sum_i^{\text{atom}} E_{\text{at}}(\mathbf{G}_i; r_{\text{cut}})$$

Fully-connected feed-forward neural network



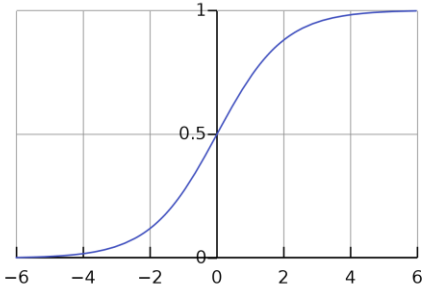


$$x_i^{l+1} = \sigma\left(\sum_j^{N^l} x_j^l w_{ij}^l + b_i^l\right)$$

x_i^l: *i*-th node in *l*-th layer
N^l: the number of nodes in *l*-th layer

Activation function (sigmoid)

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

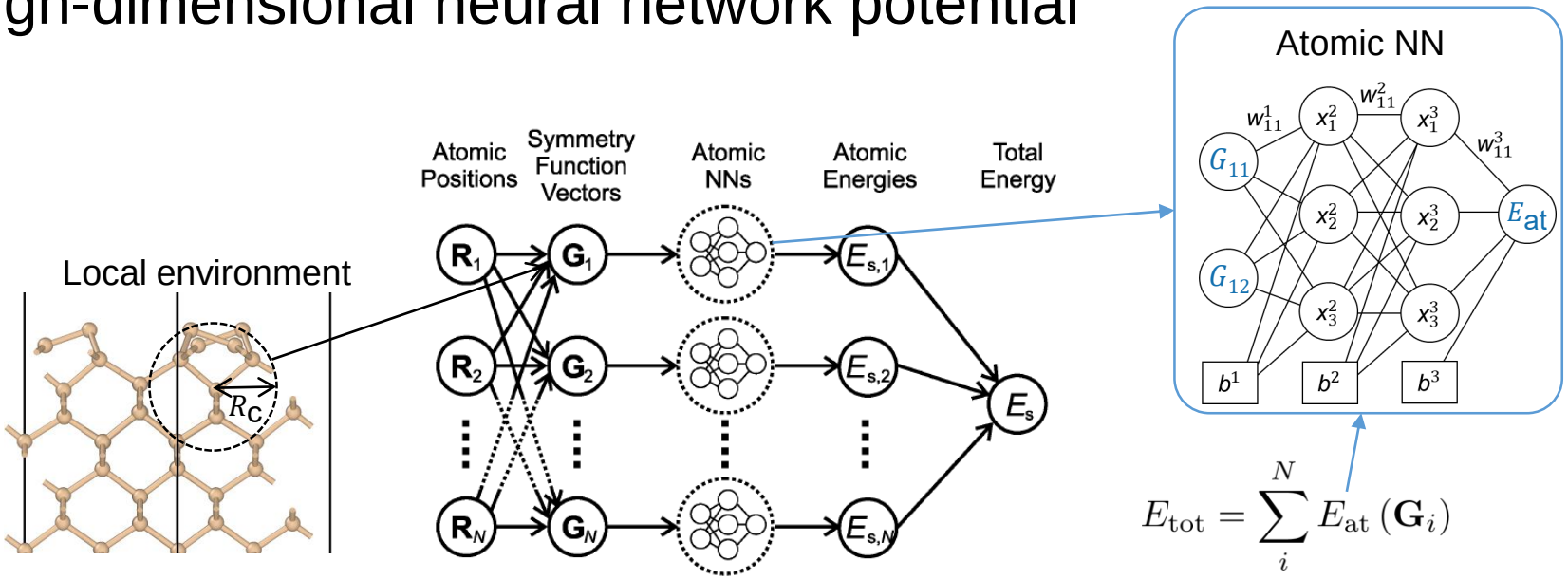


w: weight, *b*: bias

(tanh, ReLU,…)

- Feed-forward neural network consists of layers of nodes, and the nodes in adjacent layers are densely connected by weights.
- The input value propagates through hidden layer to output value.
- With non-linear activation function, NN can represent any continuous function given enough network size.

High-dimensional neural network potential

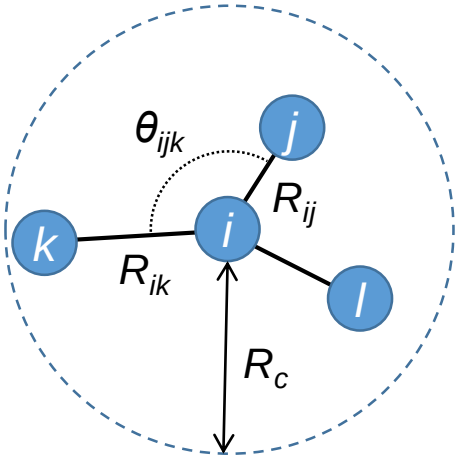


$$E_{\mathrm{tot}} = \sum_i^N E_{\mathrm{at}}\left(\mathbf{G}_i\right)$$

$$F_{k,\alpha} = -\frac{\partial E_{\mathrm{tot}}}{\partial R_{k,\alpha}} = -\sum_i^{N_{\mathrm{atom}}} \sum_s^{N_{\mathrm{SF}}} \frac{\partial E_{\mathrm{at},i}}{\partial G_{i,s}} \frac{\partial G_{i,s}}{\partial R_{k,\alpha}}$$

- For each atom, local environment is encoded into a symmetry function vector.
- Atomic NN takes symmetry function vector as input and returns atomic energy as output.
- It is assumed that the total energy is a sum of atomic energies.
- Same chemical species share the same atomic NN.

Atom-centered symmetry function



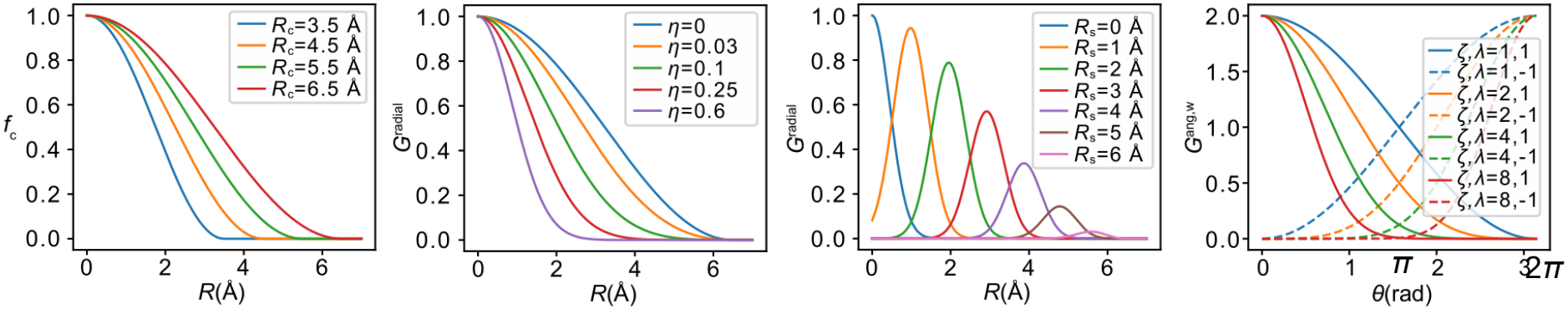
$$G_i^{\text{radial}} = \sum_j e^{-\eta(R_{ij}-R_s)^2} f_c\left(R_{ij}; R_c\right) \qquad (G_2)$$

$$f_c(R_{ij}; R_c) = \begin{cases} 0.5\left[\cos\left(\frac{\pi R_{ij}}{R_c}\right)+1\right], & \text{if } R_{ij} < R_c \\ 0, & \text{otherwise} \end{cases}$$

$$G_i^{\text{ang.w.}} = 2^{1-\zeta} \sum_{j,k\neq i} (1+\lambda\cos\theta_{ijk})^\zeta e^{-\eta\left(R_{ij}^2+R_{ik}^2\right)} f_c\left(R_{ij}; R_c\right) f_c\left(R_{ik}; R_c\right) \qquad (G_5)$$

$$G_i^{\text{ang.n.}} = 2^{1-\zeta} \sum_{j,k\neq i} (1+\lambda\cos\theta_{ijk})^\zeta e^{-\eta\left(R_{ij}^2+R_{jk}^2+R_{ik}^2\right)} \qquad (G_4)$$

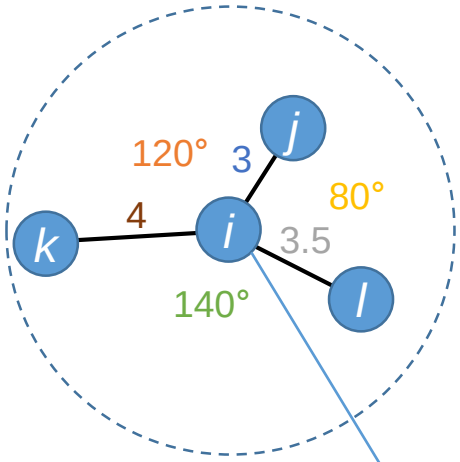
$$\times f_c\left(R_{ij}; R_c\right) f_c\left(R_{jk}; R_c\right) f_c\left(R_{ik}; R_c\right)$$



- Symmetry functions are invariant to translation and rotation of whole system, and permutation of same species.

Atom-centered symmetry function (example)

Example) parameter set consists of two radial functions with $\{\eta_1, \eta_2\}$ and two wide angular functions with $\{\zeta_1, \zeta_2\}$.



$$G_i^{\mathrm{radial},\eta_1} = e^{-\eta_1 3^2} f_c(3) + e^{-\eta_1 3.5^2} f_c(3.5) + e^{-\eta_1 4^2} f_c(4)$$

$$G_i^{\mathrm{radial},\eta_2} = e^{-\eta_2 3^2} f_c(3) + e^{-\eta_2 3.5^2} f_c(3.5) + e^{-\eta_2 4^2} f_c(4)$$

$$\begin{aligned} G_i^{\mathrm{ang,w},\zeta_1} = 2^{1-\zeta_1}\big[& (1+\lambda\cos 80)^{\zeta_1} \cdot e^{-\eta\left(3^2+3.5^2\right)} f_c(3)\, f_c(3.5) \\ & + (1+\lambda\cos 120)^{\zeta_1} \cdot e^{-\eta\left(3^2+4^2\right)} f_c(3)\, f_c(4) \\ & + (1+\lambda\cos 140)^{\zeta_1} \cdot e^{-\eta\left(3.5^2+4^2\right)} f_c(3.5)\, f_c(4)\big] \end{aligned}$$

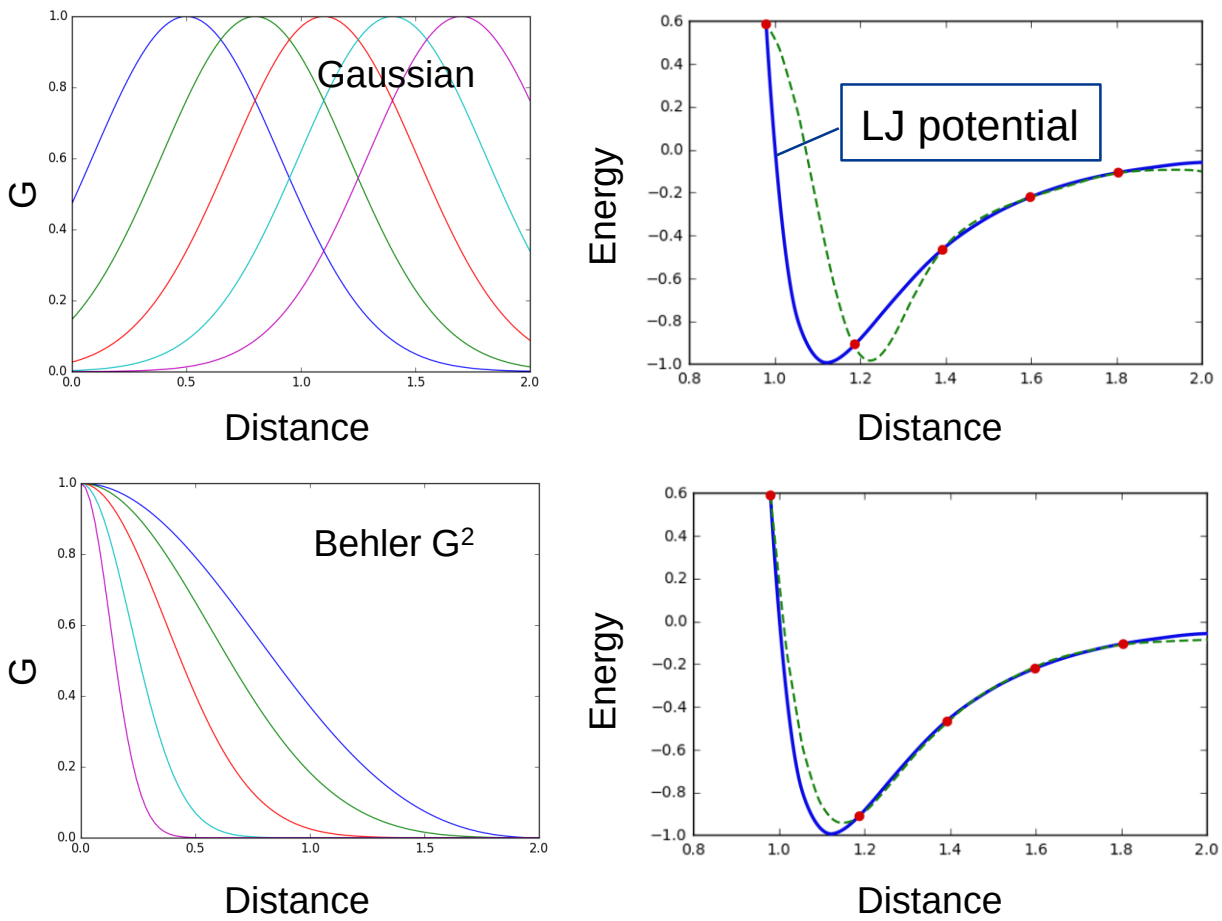
$$\begin{aligned} G_i^{\mathrm{ang,w},\zeta_2} = 2^{1-\zeta_2}\big[& (1+\lambda\cos 80)^{\zeta_2} \cdot e^{-\eta\left(3^2+3.5^2\right)} f_c(3)\, f_c(3.5) \\ & + (1+\lambda\cos 120)^{\zeta_2} \cdot e^{-\eta\left(3^2+4^2\right)} f_c(3)\, f_c(4) \\ & + (1+\lambda\cos 140)^{\zeta_2} \cdot e^{-\eta\left(3.5^2+4^2\right)} f_c(3.5)\, f_c(4)\big] \end{aligned}$$

Symmetry function vector $\mathbf{G}_i$ for atom i

$$\mathbf{G}_i = \left[G_i^{\mathrm{radial},\eta_1}, G_i^{\mathrm{radial},\eta_2}, G_i^{\mathrm{ang,w},\zeta_1}, G_i^{\mathrm{ang,w},\zeta_2}\right]$$

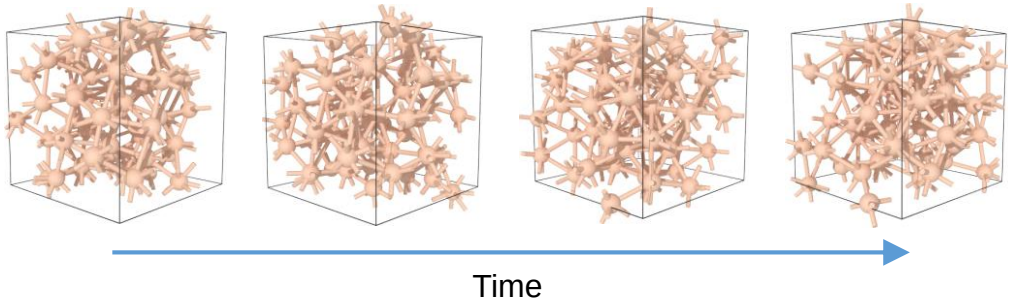
- Symmetry functions with different parameters together compose a vector, which describes the local environment → symmetry function vector

Importance of descriptors



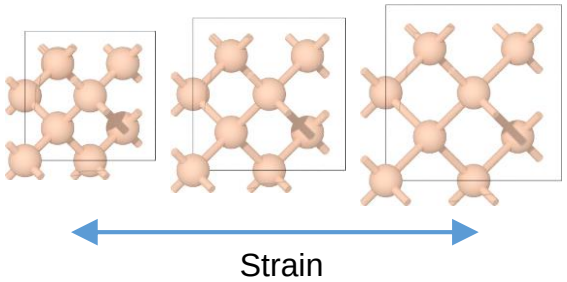
Training set

Simulation trajectory (e.g. molecular dynamics)



OR

Procedurally generated structures
(e.g. strain, random displacement)



For selected snapshots

$$
\begin{array}{lllll}
\mathbf{G}_1 & = [G_1^1, & G_1^2, & G_1^3, & \cdots, G_1^{N_{\mathrm{SF}}}] \\
\mathbf{G}_2 & = [G_2^1, & G_2^2, & G_2^3, & \cdots, G_2^{N_{\mathrm{SF}}}] \\
\vdots & \vdots & \vdots & \vdots & \ddots \quad \vdots \\
\mathbf{G}_{N_{\mathrm{atom}}} & = [G_{N_{\mathrm{atom}}}^1, & G_{N_{\mathrm{atom}}}^2, & G_{N_{\mathrm{atom}}}^3, & \cdots, G_{N_{\mathrm{atom}}}^{N_{\mathrm{SF}}}]
\end{array}
$$

E_{tot} (total energy)

F (atomic forces)

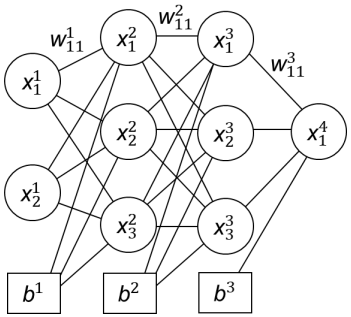
S (stress)

Input descriptors (for all atoms)

Output properties from DFT

Training process

$$w^l \to w^l - \alpha \frac{\partial \Gamma}{\partial w^l}$$



- Training is the process of minimizing loss function by updating weights.
- Loss function is defined as the error between training set and the prediction.

Energy loss functions

$$\Gamma_E = \frac{1}{M} \sum_{i=1}^{M} \left(\frac{E_i^{\mathrm{DFT}} - E_i^{\mathrm{NNP}}}{N_i} \right)^2$$

$$\Gamma'_E = \frac{1}{M} \sum_{i=1}^{M} \left(E_i^{\mathrm{DFT}} - E_i^{\mathrm{NNP}} \right)^2$$

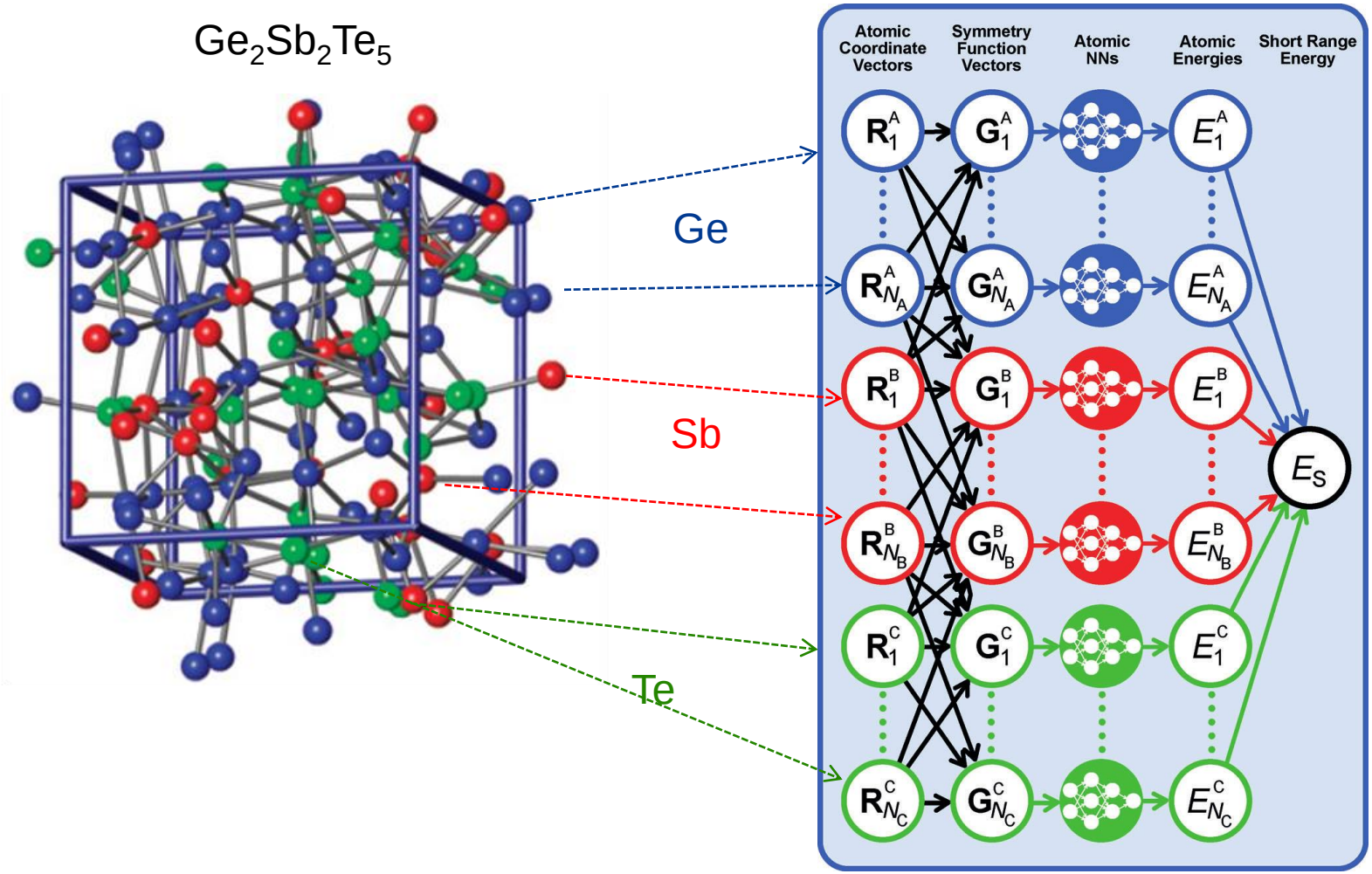
Force loss functions

$$\Gamma_F = \frac{1}{3\sum_{i=1}^{M} N_i} \sum_{i=1}^{M} \sum_{j=1}^{N_i} \left| \mathbf{F}_{ij}^{\mathrm{DFT}} - \mathbf{F}_{ij}^{\mathrm{NNP}} \right|^2$$

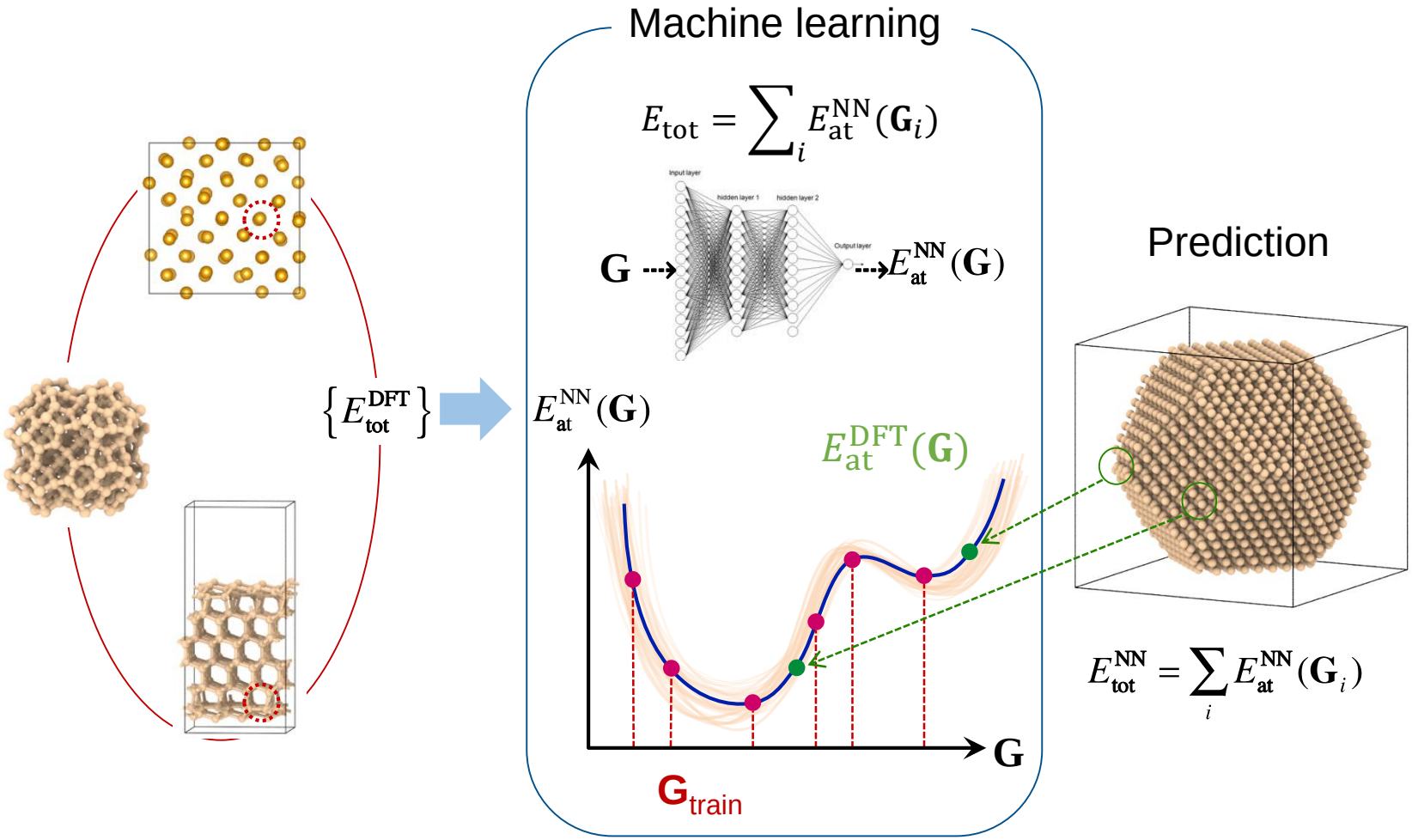
$$\Gamma'_F = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{3N_i} \sum_{j=1}^{N_i} \left| \mathbf{F}_{ij}^{\mathrm{DFT}} - \mathbf{F}_{ij}^{\mathrm{NNP}} \right|^2$$

$$\Gamma = \frac{1}{M} \sum_{i=1}^{M} \left(\frac{E_i^{\mathrm{DFT}} - E_i^{\mathrm{NNP}}}{N_i} \right)^2 + \frac{\mu}{3\sum_{i=1}^{M} N_i} \sum_{i=1}^{M} \sum_{j=1}^{N_i} |\mathbf{F}_{ij}^{\mathrm{DFT}} - \mathbf{F}_{ij}^{\mathrm{NNP}}|^2 + \frac{\nu}{6M} \sum_{i=1}^{M} \sum_{k=1}^{6} |\mathbf{S}_{ik}^{\mathrm{DFT}} - \mathbf{S}_{ik}^{\mathrm{NNP}}|^2$$

For multicomponent systems, one NN for each type. Symmetry functions separate contributions from different types. Therefore, G_2 scales with N_{type} and G_4 scales with $\frac{1}{2}(N_{type})(N_{type}-1)$.



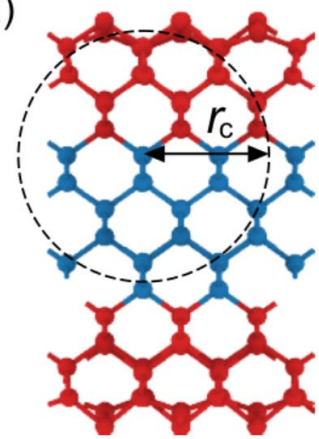
Neural network potentials



- The machine-learning potential is essentially a variant of O(*N*) method for DFT calculations, which utilizes the atomic energy.

Atomic energy mapping

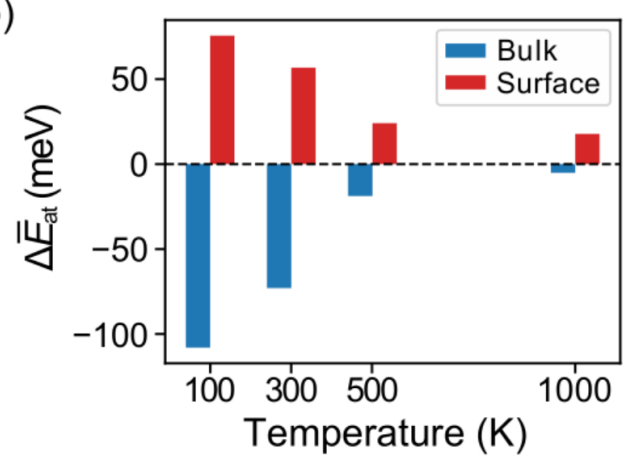
(a)



$$\bar{E}_{\mathrm{at}}^{\mathrm{DFT}} = \left[E_{\mathrm{tot}}^{\mathrm{DFT}} - N_b \cdot E_{\mathrm{at}}^{\mathrm{DFT}}(\mathrm{bulk})\right]/N_s$$

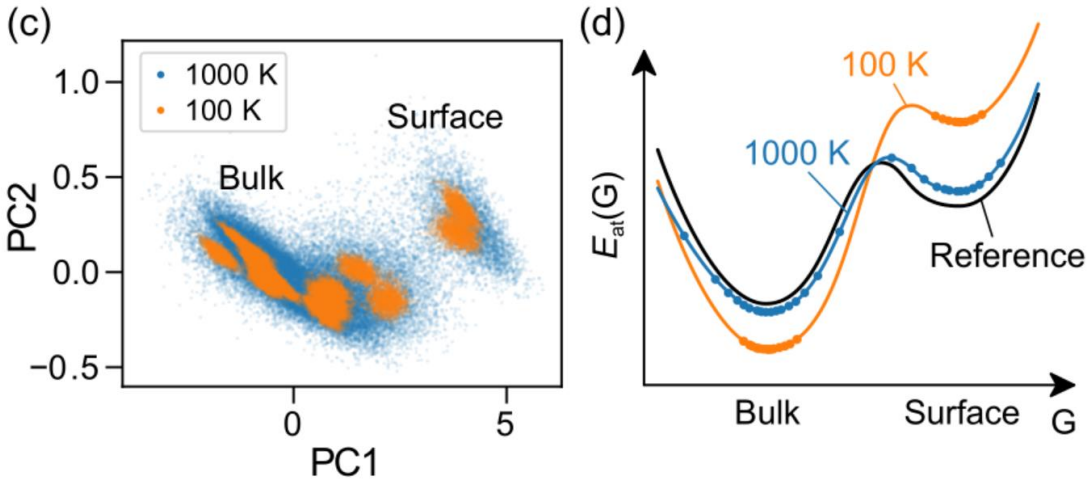
$$\bar{E}_{\mathrm{at}}^{\mathrm{DFT}} = E_{\mathrm{at}}^{\mathrm{DFT}}(\mathrm{bulk})$$

(b)



- It is possible for NNP to give accurate total energies for training set but give incorrect atomic energy function (ad hoc energy mapping).
- Training set consists of MD snapshots of Si slab at various temperatures.
- The average atomic energy errors for bulk region and surface region are calculated.
- At low temperature, the atomic energy error is large, but the error decreases as the temperature of training set increases.

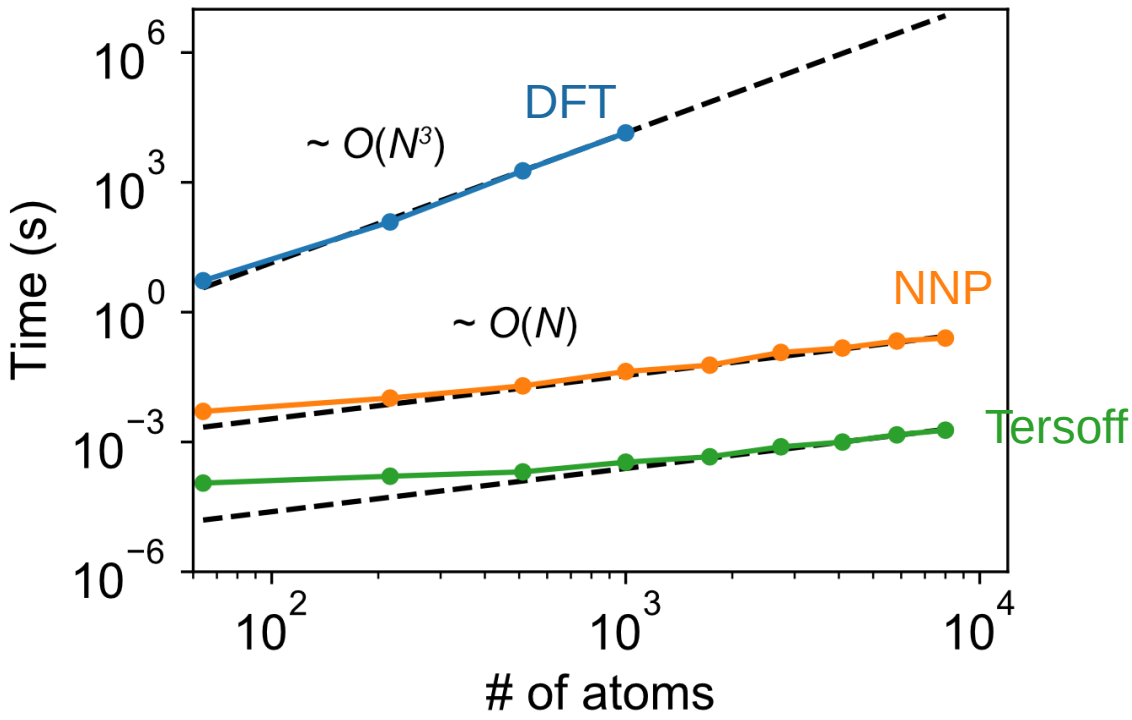
Atomic energy mapping



- This is because the samples are disconnected at low temperature, but more connected at high temperature.
- When configurations are disconnected, the relative offset between configurations can vary while the total energy remains constant.

Computational performance of NNP

Ex) Crystalline Si

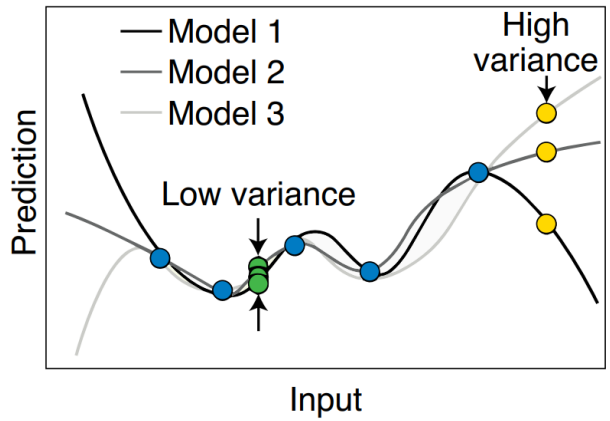


✓ Machine learning potential allows for MD simulations at the DFT accuracy up to 100,000 atoms or 10-nm feature size over the time scale of nanoseconds.

Error quantification

It is recommended that errors are monitored by uncertainty estimation during MD.

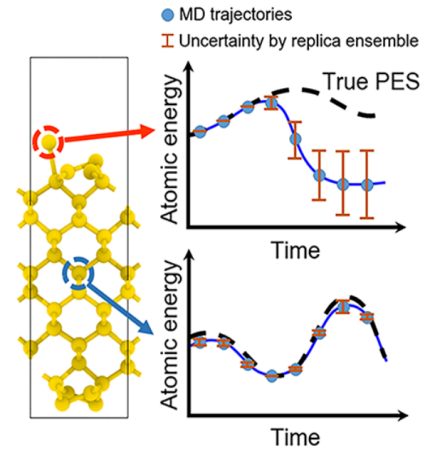
Ensemble of NNP



P. Friederich *et al.*, *Nat. Mater.* **20**, 750–761 (2021).

- An ensemble of NNP shows high variance in untrained domain.
- It is difficult to localize atoms with high uncertainty.

Replica ensemble



W. Jeong, *et al.*, J. Phys. Chem. Lett., **11**, 6090−6096 (2020).

- Replica ensemble is trained on atomic energy of the reference NNP.
- Only problematic atoms show high variance in atomic energy.

II. 실전 이론

Four technical elements in MLP application:

- Training set
- Feature engineering
- Network dimension
- Training

The machine learning potential is useful only when:

i) it is sufficiently accurate,

ii) the target simulation is meaningful and well beyond DFT capability,

iii) building the training set is not computationally too heavy.

Comparison of various MLP codes

Code name	Potential type	Model	Descriptor	Reference
SIMPLE-NN	Neural Network Potential	Neural Network	BP	Computer Physics Communications 242 (2019) 95–103
Amp			BP Zernike descriptor Bispectrum	Computer Physics Communications 207 (2016) 310–324
Aenet			Chebyshev polynomial with angular, radial distribution functions	J. Chem. Phys. 155, 074801 (2021)
DeepMD-kit			Radial, angular descriptor	Computer Physics Communications 228 (2018) 178–184
ANI-1			Heavily modified BP	Chem. Sci. (2017) 8, 3192–3203
PROPhet			BP	Sci. Rep. (2017) 7, 1192
ruNNer			BP	ACS Catal. 2017, 7, 6600−6608
QUIP	Gaussian Approxmation Potential	Kernel based	Smooth overlap of atomic positions (SOAP)	International Journal of Quantum Chemistry 2015, 115, 1051–1057
MLIP	Moment Tensor Potential	Polynomial	Moment tensor descriptor	Mach. Learn.: Sci. Technol. 2 (2021) 025002

SIMPLE-NN

SNU Interatomic Machine-learning PotentiaL packagE – version Neural Network

github.com/MDIL-SNU/SIMPLE-NN

K. Lee et al, Comp. Phys. Comm. 242, 95 (2019)

DFT training sets:
$\{E_{tot}, \mathbf{F}_i, \boldsymbol{\sigma}\}$
from VASP,
Quantum-Espresso,

ASE →

SIMPLE-NN

Version 2.0

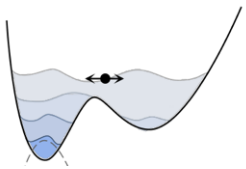
Interfacing module →

LAMMPS

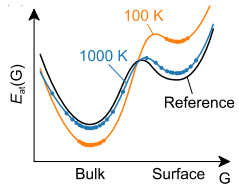
- Generation of G vectors
- Input preprocessing supports MPI
- Training via PyTorch (supports GPU)
- PCA & whitening
- Dropout for regularization
- Uniform training to rectify sample bias (JPCC **122**, 22790 (2018)).
- Replica ensemble for uncertainty estimation (JPCL **11**, 6090 (2020)).

PyTorch

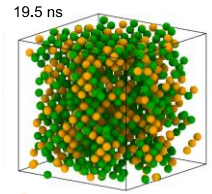
Metadynamics sampling



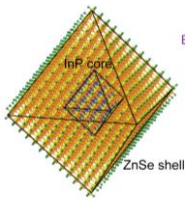
Atomic energy mapping



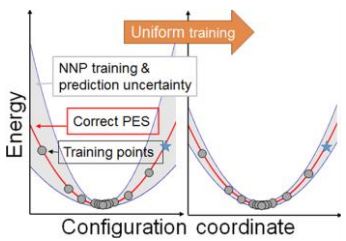
GeTe crystallization



InP quantum dot



GDF weighting

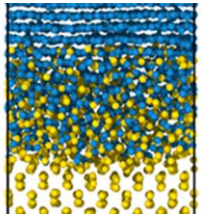


SIMPLE-NN

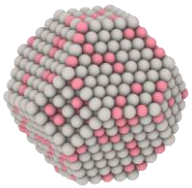
github.com/MDIL-SNU/SIMPLE-NN

- J. Phys. Chem. C 122, 22790 (2018)
- Comp. Phys. Comm. 242, 95 (2019)
- Phys. Rev. Matt. 3, 093802 (2019)
- Comp. Mat. Sci. 181, 109725 (2020)
- J. Phys. Chem. Lett. **11**, 6090 (2020) Suppl. Cover
- Phys. Rev. B **102**, 224104 (2020)
- npj Computational Materials (2021)
- arXiv:2107.02594
- ACS Materials Au (2021) Suppl. Cover

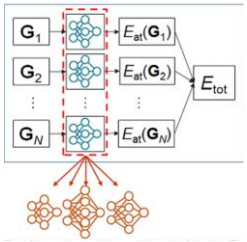
Ni silicidation



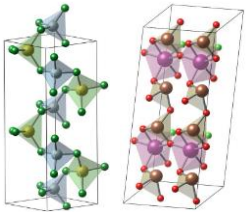
Pt_3Co catalysts



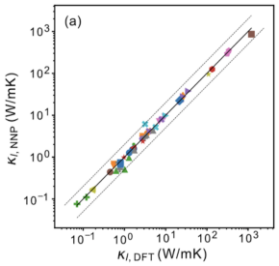
Uncertainty quantification



Crystal structure prediction



Thermal conductivity



Four technical elements in MLP application:

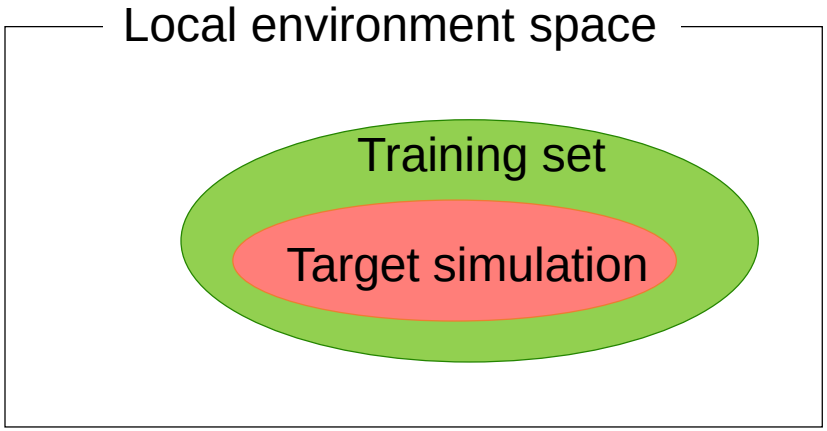
- Training set
- Feature engineering
- Network dimension
- Training

Training set

Machine learning potential = data-driven DFT calculation

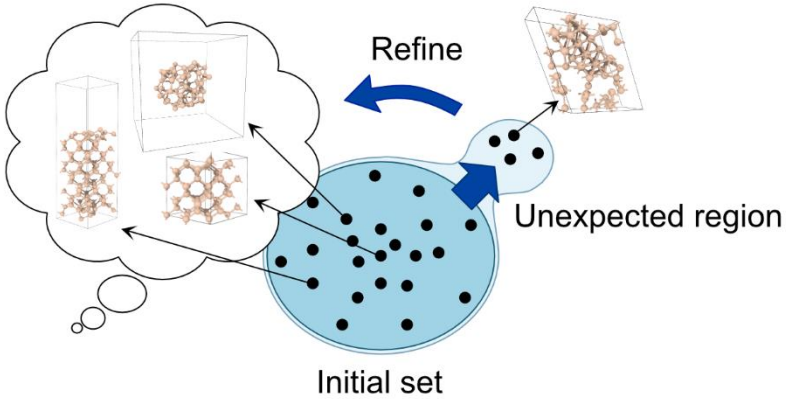
Data is everything!

The success of the potential depends on whether one can find a proper training set.



Training set =

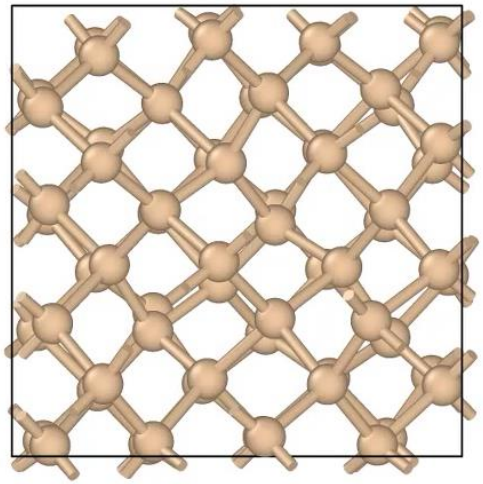
Basic type + application type

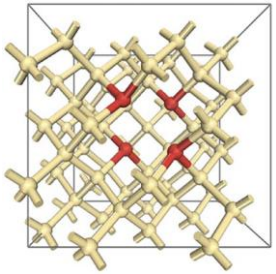


- Constructing training set depends mostly on human intuition and repeated refinement.
- This is a major bottleneck against wide use of MLPs.

Example: Si

Target simulation: MD simulation of pure Si at 1000 K with 512 atoms (4x4x4 supercell).



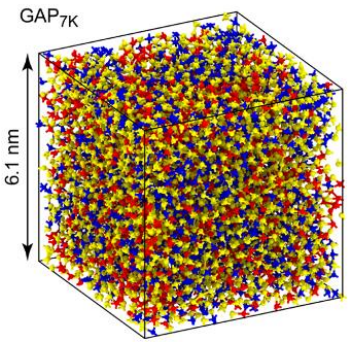


64 atoms (2x2x2 supercell)
1-ps MD @ 1300 K
sampled every 10 fs.

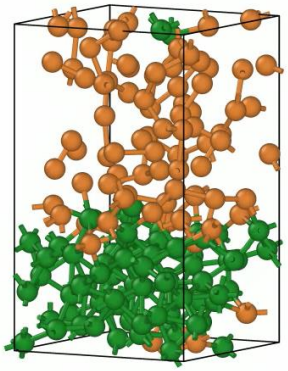
+ Equation of states, lattice distortions for elastic properties.

NVT: recommended to include stress tensors into loss function.
NPT: may need a higher accuracy than NVT.

Example: GeTe

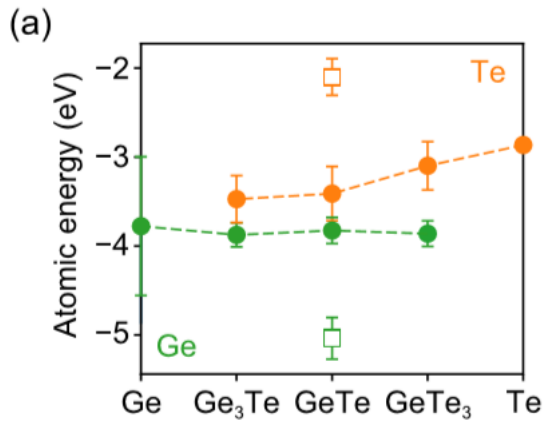


Data description	# structures	# data points	Energy RMSE (meV/atom)	Force RMSE (eV/Å)
GeTe				
Strained FCC	61	488	3.45	0.14
FCC crystal MD	125	8,000		
Strained HEX	102	612	3.22	0.17
HEX crystal MD	125	13,500		
Melting of crystal (FCC)	352	33,792	4.65	0.24
Liquid @1000K	500	48,000	5.00	0.26
Quenching (1000-300K)	518	49,728		
Amorphous @500K	251	24,096	3.55	0.26
Ge				
Liquid @1000K	125	12,500	3.09	0.17
Te				
Liquid @1000K	250	25,000	4.83	0.30
etc.				
Interface reaction	240	38,400	3.97	0.27
	61	13,420		



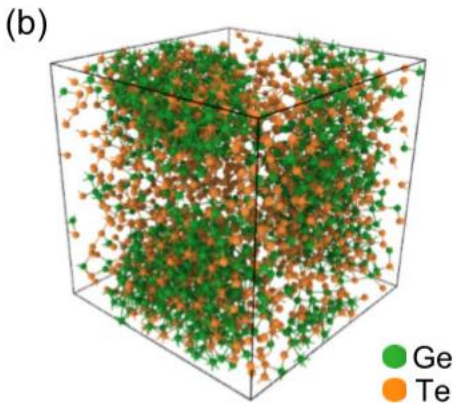
D. Lee and K. Lee et al, Computational Materials Science **181**, 109725 (2020)

Ad hoc energy mapping in multi-component system



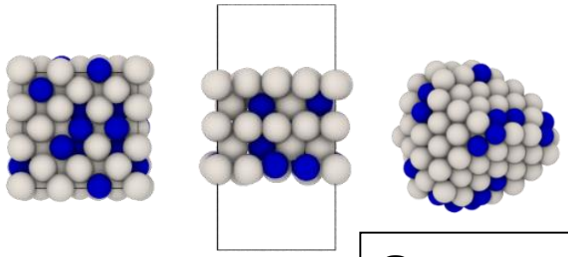
□ Training set includes only 1:1

● Training set includes whole composition range



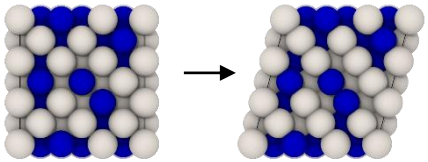
- In multi-component systems, additionally relative offset between chemical species can lead to ad hoc energy mapping.
- The ad hoc energy mapping between chemical species undermines stability of NNP.
- For example, ad hoc energy mapping caused phase separation in GeTe liquid.
- The problem was solved by sampling continuously over the composition range.

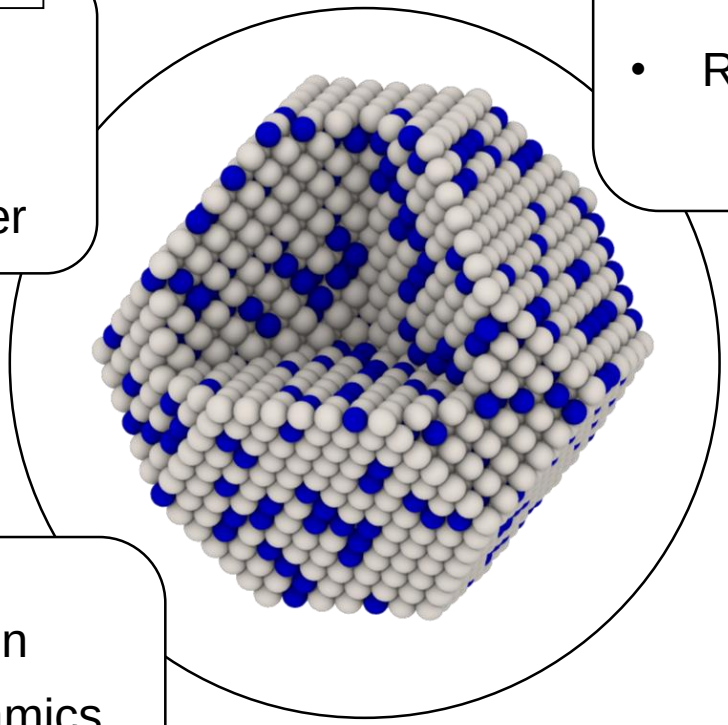
Example: Pt_3Co nanocluster



Structure

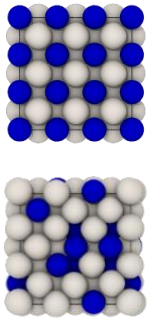
- Bulk
- Slab
- Cluster





Composition

- Intermetallic compound
  : Pt, Pt_3Co, PtCo, Co
- Random alloy ($Pt_{1-x}Co_x$)
  : $0.1 \leq x \leq 0.4$



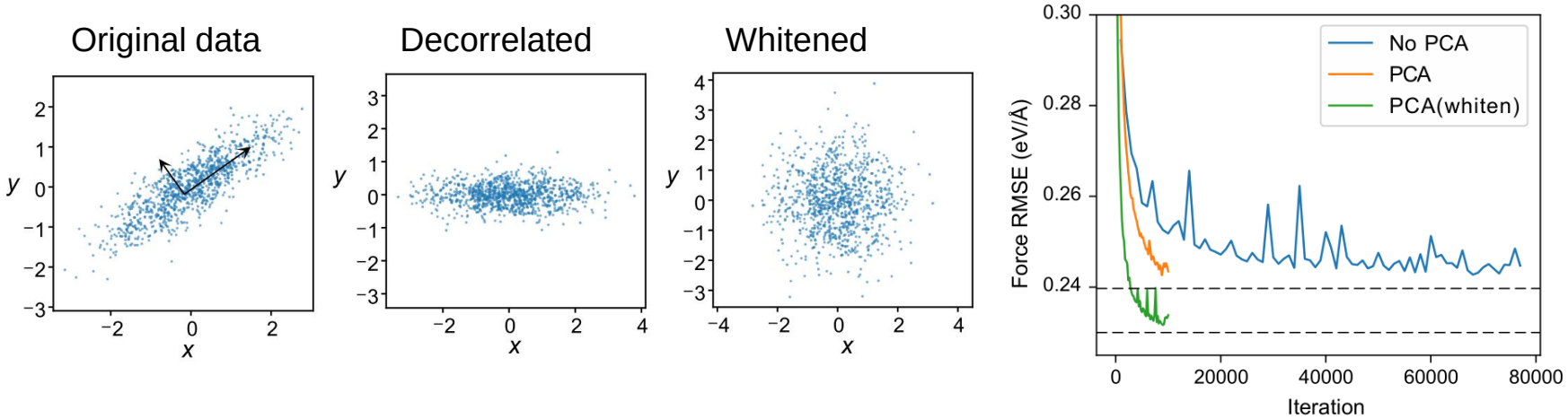
Variation

- Lattice distortion
- Molecular dynamics
- Vacancy formation
- Vacancy diffusion

Accuracy of training set

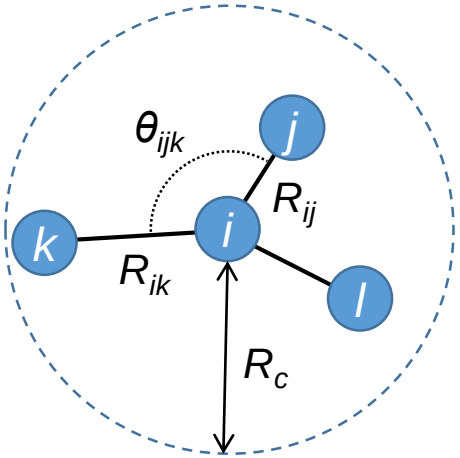
Since the DFT-MD simulation takes a long time, one solution is to carry out DFT-MD with low accuracy and recalculate sampled snapshots with high or normal accuracy.

Input preprocessing



- For efficient training, input preprocessing is recommended.
- We scale each symmetry function component so that all components are in the similar range.
- Symmetry functions are highly correlated. High correlation between inputs slows down the training process.
- Preprocessing inputs with principal component analysis (PCA) improves convergence significantly.

Atom-centered symmetry function



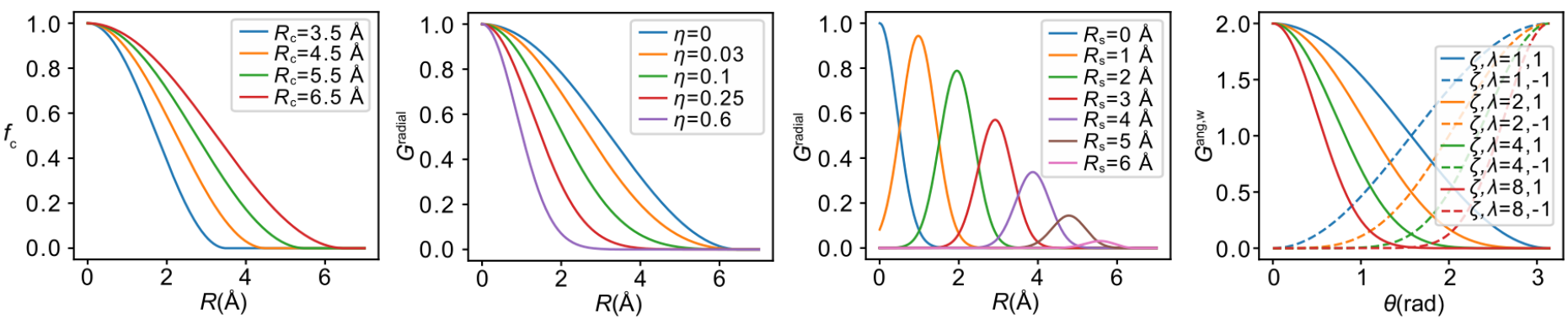
$$G_i^{\text{radial}} = \sum_j e^{-\eta(R_{ij}-R_s)^2} f_c\left(R_{ij}; R_c\right) \quad \text{(G2)}$$

$$f_c(R_{ij}; R_c) = \begin{cases} 0.5\left[\cos\left(\frac{\pi R_{ij}}{R_c}\right)+1\right], & \text{if } R_{ij} < R_c \\ 0, & \text{otherwise} \end{cases}$$

$$G_i^{\text{ang.w.}} = 2^{1-\zeta} \sum_{j,k\neq i} (1+\lambda\cos\theta_{ijk})^\zeta e^{-\eta\left(R_{ij}^2+R_{ik}^2\right)} f_c\left(R_{ij}; R_c\right) f_c\left(R_{ik}; R_c\right) \quad \text{(G5)}$$

$$G_i^{\text{ang.n.}} = 2^{1-\zeta} \sum_{j,k\neq i} (1+\lambda\cos\theta_{ijk})^\zeta e^{-\eta\left(R_{ij}^2+R_{jk}^2+R_{ik}^2\right)} \quad \text{(G4)}$$

$$\times f_c\left(R_{ij}; R_c\right) f_c\left(R_{jk}; R_c\right) f_c\left(R_{ik}; R_c\right)$$



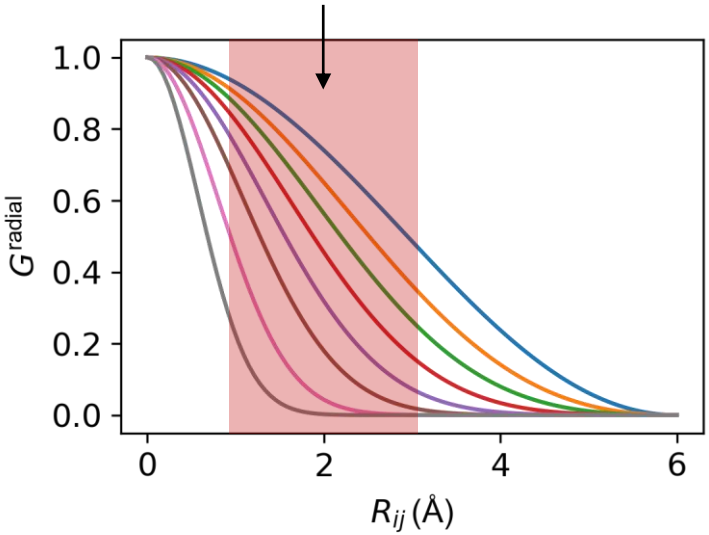
- Symmetry functions are invariant to translation and rotation of whole system, and permutation of same species.

Selection of parameters for symmetry functions

- We commonly use the 8 radial (G2) and 18 angular (G4) symmetry functions for each type of pair and triplet with the parameters reported in the previous research* (fitted to TiO_2).

- The parameters are selected to detect best the change of local configuration.

The steepest part of G^{radial}
= the most sensitive region



η (Å^{-2}) with R_s of 0	
0.003214	0.214264
0.035711	0.357106
0.071421	0.714213
0.124987	1.428426

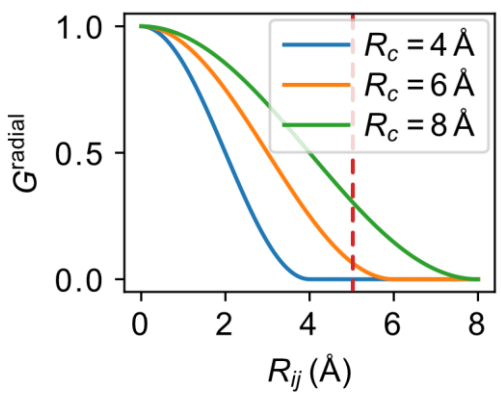
Comput. Mater. Sci. **114**, 135–150 (2016)

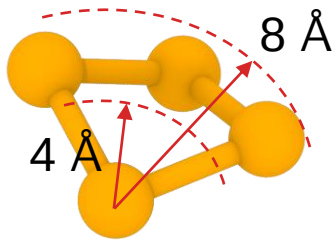
- In general, η < 0.09, λ of {1, -1}, and ζ of {1, 2, 4} are used in $G^{angular}$.

Cutoff radius

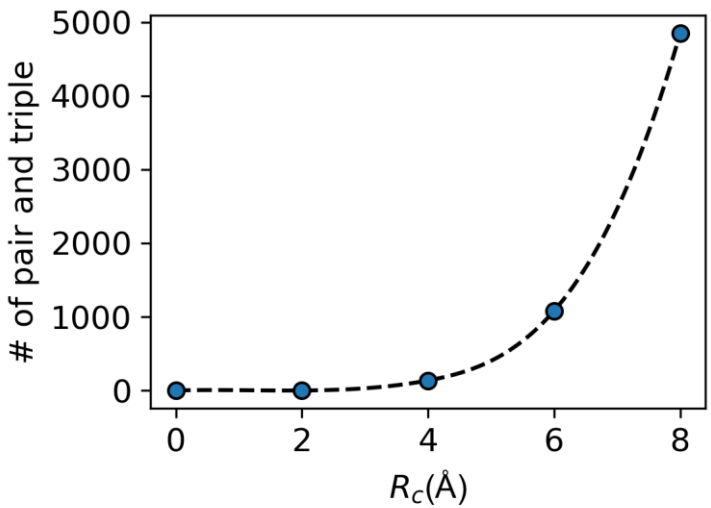
$$G_i^{\text{radial}} = \sum_j e^{-\eta(R_{ij}-R_s)^2} f_c\left(R_{ij}; \boxed{R_c}\right)$$

$$f_c(R_{ij}; R_c) = \begin{cases} 0.5\left[\cos\left(\frac{\pi R_{ij}}{R_c}\right)+1\right], & \text{if } R_{ij} < R_c \\ 0, & \text{otherwise} \end{cases}$$



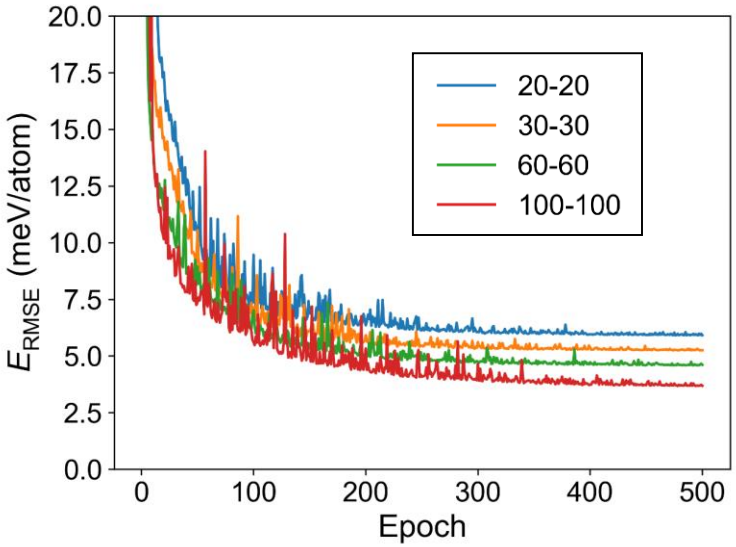


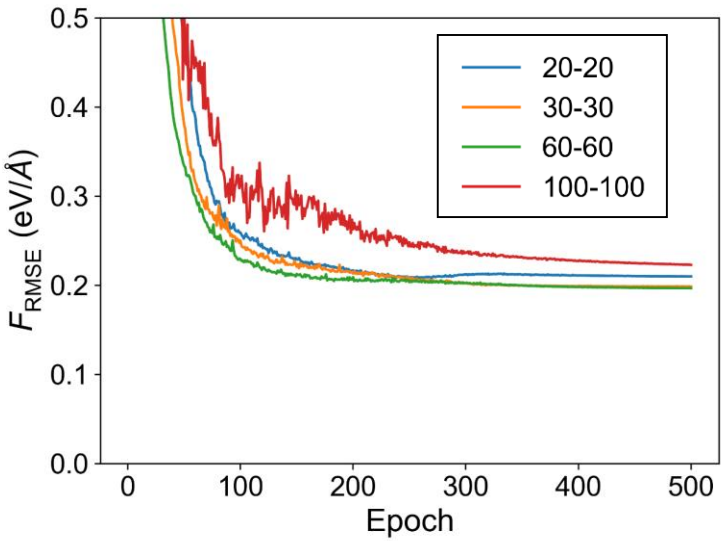
The number of pair and triplet

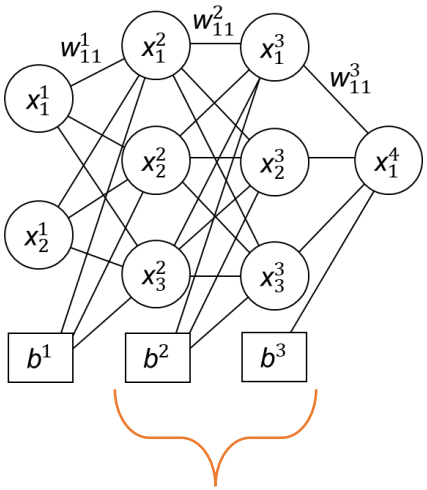


- Sometimes, practitioner should modify the parameters of input feature for high accuracy at a certain distance or angle.
- The computational cost increases rapidly with the cutoff distance.

Dimension of neural network

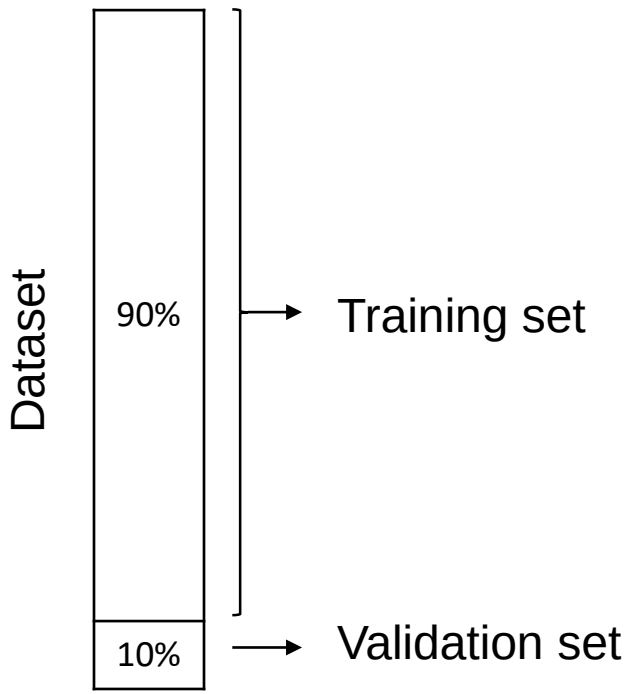




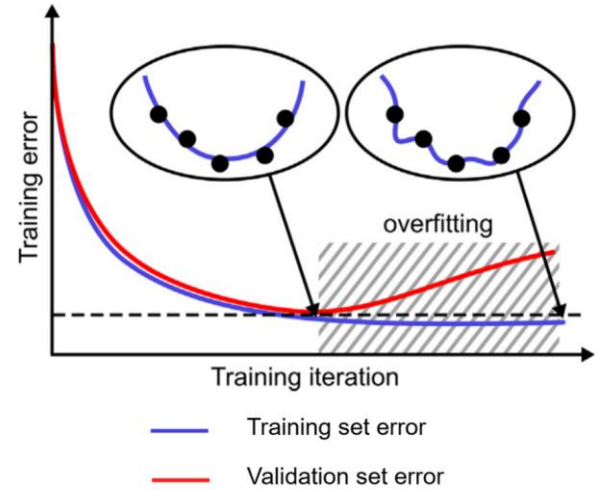


- 대부분의 경우 layer의 수는 2~3 개, node의 수는 30~70으로 설정함.
- 더 높은 정확도가 필요한 경우 symmetry function vector의 길이를 늘리면서 hidden layer도 더 크고 깊게 설정할 수도 있음.

Training, validation sets



- Overfitting*

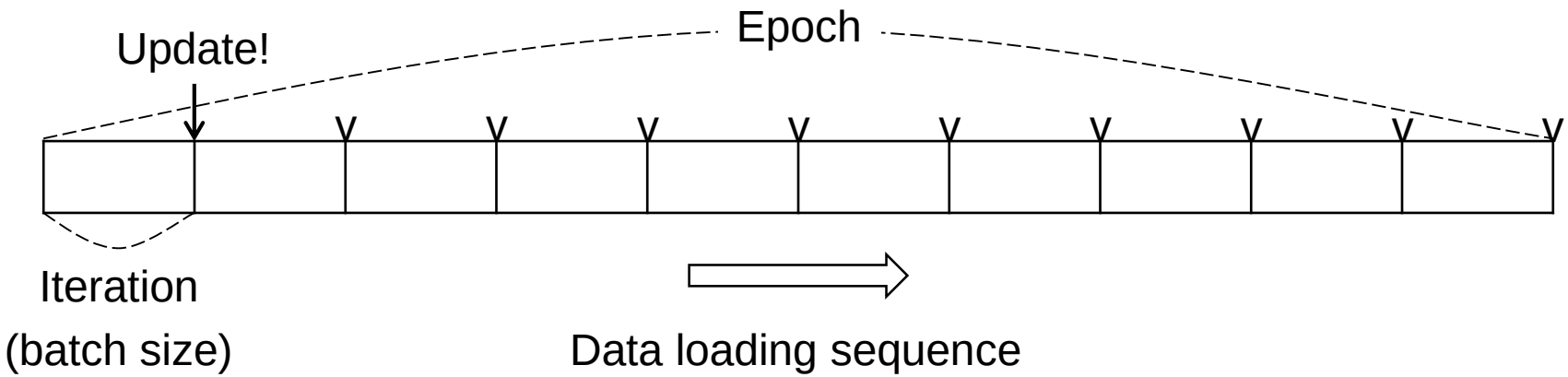


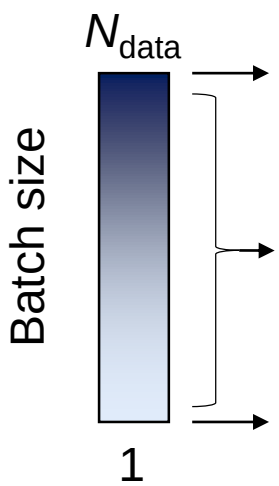
- As network is optimized to minimize the training error, the validation set error should be monitored.

	Network update	Error check during training	Purpose
Training	O	O	To update the network
Validation	X	O	To monitor the overfitting*

Batch size

Batch size → Size of data in updating weights

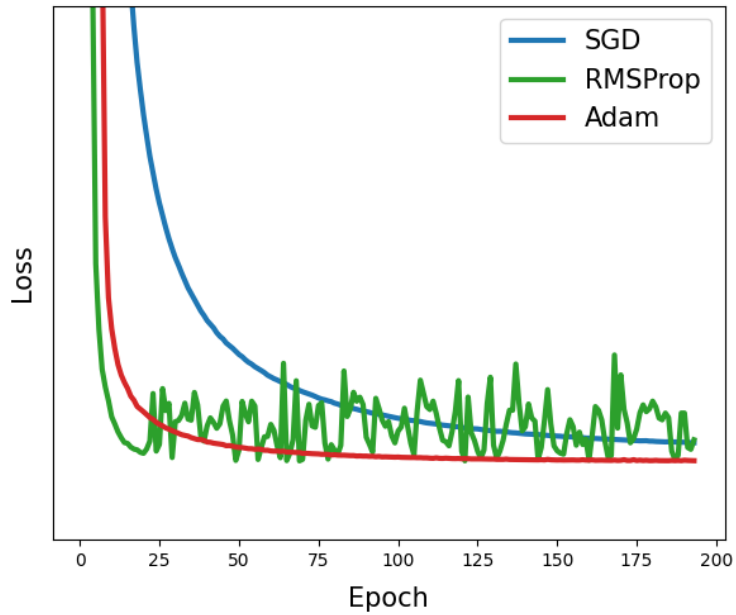




	Memory	Convergence
• Batch gradient descent • Full batch	Large	Good
• Mini batch gradient descent	Medium	Moderate
• Stochastic gradient descent • Online learning	Small	Poor

Optimizer

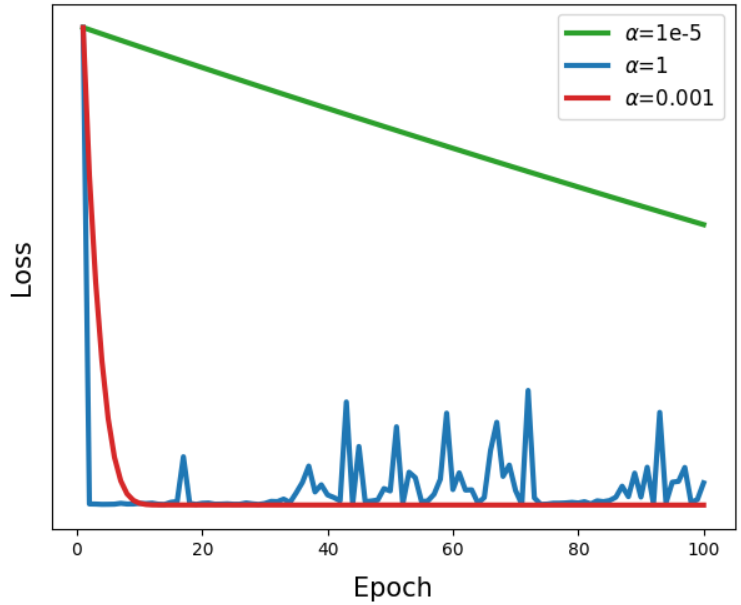
Optimizer



- Algorithm to update weights
- Adam optimizer generally works well

Learning rate

$$w^l \to w^l - \alpha \frac{\partial \Gamma}{\partial w^l}$$



- The decay rate and convergence tendency of loss function depend on the learning rate

Loss function

0.1-1

10^{-6}

$$\Gamma = \frac{1}{M}\sum_{i=1}^{M}\left(\frac{E_i^{\mathrm{DFT}} - E_i^{\mathrm{NNP}}}{N_i}\right)^2 + \frac{\mu}{3\sum_{i=1}^{M} N_i}\sum_{i=1}^{M}\sum_{j=1}^{N_i}|\mathbf{F}_{ij}^{\mathrm{DFT}} - \mathbf{F}_{ij}^{\mathrm{NNP}}|^2 + \frac{\nu}{6M}\sum_{i=1}^{M}\sum_{k=1}^{6}|\mathbf{S}_{ik}^{\mathrm{DFT}} - \mathbf{S}_{ik}^{\mathrm{NNP}}|^2$$

Loss function with L2 regularization:

$$\Gamma_{L2} = \Gamma + \lambda\{\sum_{l=0}^{\text{Layer}}\sum_{i,j=0}^{\text{Nodes}}|w_{ij}^{l}|^2 + \sum_{l=0}^{\text{Layer}}\sum_{i=0}^{\text{Nodes}}\left|b_i^{l}\right|^2\}$$

Analysis of errors

- 학습 후 E_{RMSE} ≤ 5~10 meV/atom, F_{RMSE} ≤ 0.2~0.3 eV/Å, S_{RMSE} ≤ ~10 kB 일 경우 보통 문제 없는 quality임.
- 목표로 하는 분자동역학 모사 과정과 관련된 기본적인 물성에 대하여 오차가 5 % 이하일 경우 괜찮은 quality임 (목표로 하는 정확도나 물성에 따라 다름).

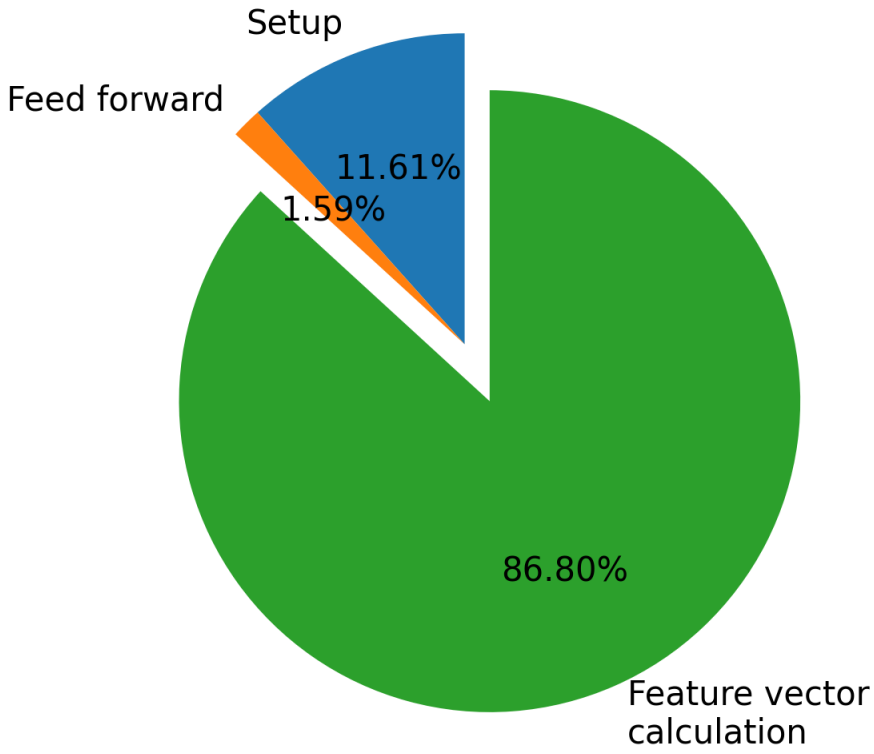
  Ex)

  i) Elastic properties for amorphous simulation

  ii) Surface energy for slab simulation

Computational cost in MD

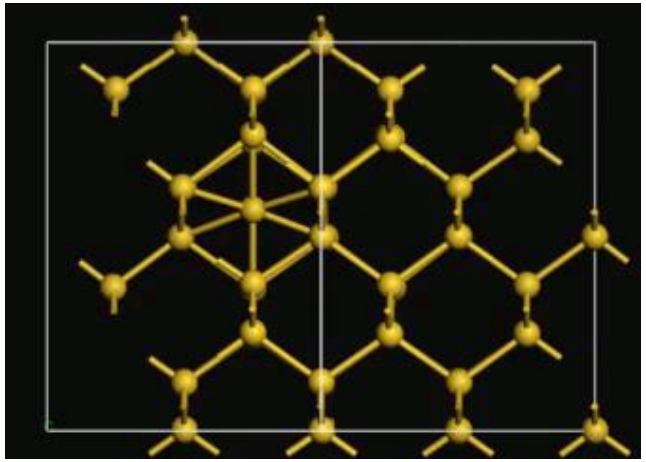
Si 1000 atom MD during 10 timestep (network: 26×30×30×1)



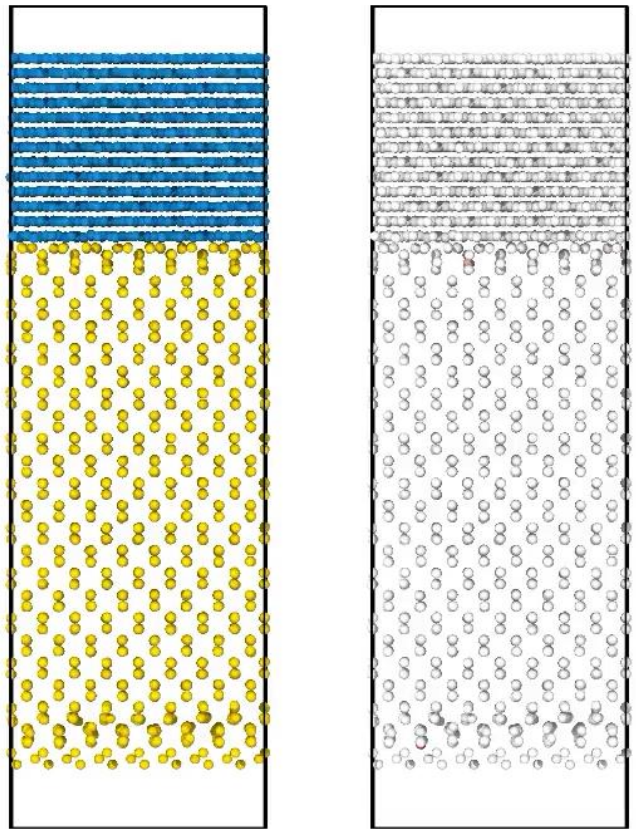
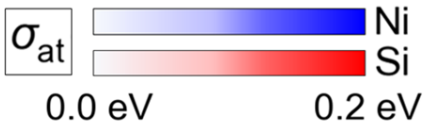
- LAMMPS spends most time (86.8%) calculating feature vectors.

Reliability of simulation

- Catastrophic failure



- Untrained domain



Uncertainty estimation
by replica ensemble